Better Late Than Never: *Late Presenting Double Jeopardy AMI-CS Successfully Treated With Multivessel PCI*

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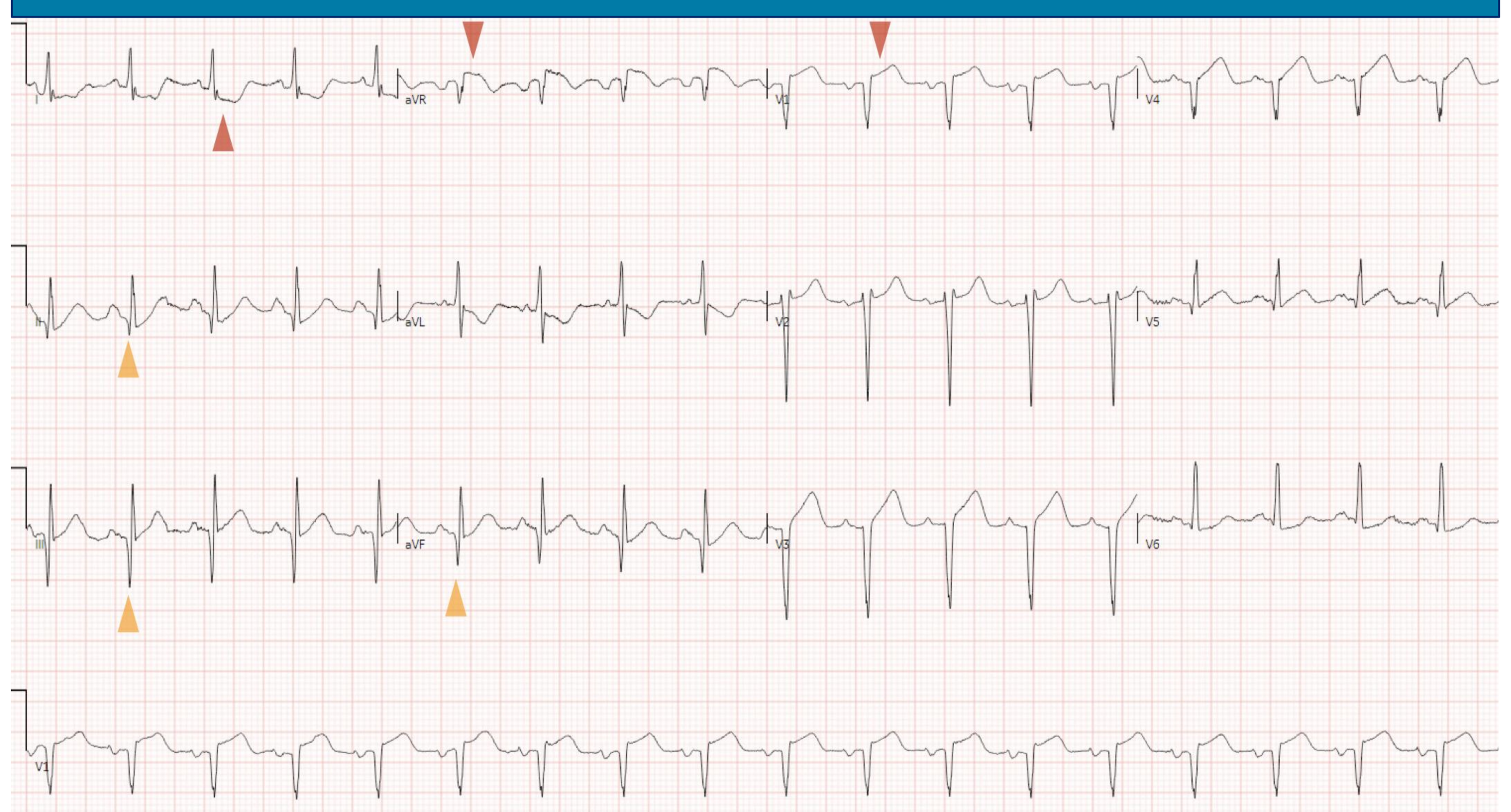
INTRODUCTION

- Acute myocardial infarct cardiogenic shock (AMI-CS) carries ~40–50% in-hospital mortality despite modern reperfusion capabilities
- A "double jeopardy myocardial infarct" occurs when a second myocardial territory (i.e., collateralized) suffers an ischemic insult in addition to the culprit lesion's native distal myocardial territory

CASE PRESENTATION

- 77 F, PMH only HTN; 14h substernal CP, radiating to back and epigastrium; a/w dyspnea, diaphoresis
- hs-Tn 4,484; NT-proBNP 6,259 pg/mL; LA 1.5; AST 118, ALT 78 U/L; Cr 0.7; WBC 16.3

PRESENTING ECG



Acute ST elevations V1, aVR with reciprocal depressions in I; deep Q-waves in inferior leads c/f chronic ischemia

Emergent Coronary Angiography & RHC

- Distal LM/ostial LAD subtotal occlusion [suspected culprit]; large first septal perforator bifurcation lesion; diffuse distal disease
- Proximal LCx subtotal occlusion; large tandem OMs
- RCA CTO; prominent L->R, R->R collaterals

CICU Management

- Early BiV AMI-CS **[SCAI B]**
- Active infusions: heparin, furosemide, nitroprusside
- Cardiac surgery consulted, adamantly declined CABG
- IABP placed → improved respiratory status; ticagrelor loaded
- TTE: EF 20–25%; septal akinesis, c/f completed LAD infarct; anterolateral dyskinesia

RHC

RA	20
RV	59/15
PA (mean)	66/44 (56)
PCWP	41
LVEDP	36
CO/CI (Fick)	2.6 / 1.7
CO/CI (TD)	2.4 / 1.6
PVR	6.0 WU
Sys BP (MAP)	96/63 (74)
CPO	0.41 W
PAPi	1.1

HIGH-RISK PCI

Preprocedural planning

- Target priority: LCx >> RCA > LAD
- Upfront Impella CP MCS, elective intubation, femoral CVL IABP; pre-closure for large-bore access

Successful multivessel PCI

- **LCx:** POBA → [LAD PCI] → LM–LCx DES
- **LAD:** POBA + DCB (stent-sparing)
- **RCA:** Antegrade CTO traversal; overlapping DES

Outcomes

- Extubated POD 1 · floor POD 2 · **home POD 4**
- TTE: EF **20–25% → 30–35%**; similar WMA pattern
- Future viability imaging ± staged LAD revascularization

CORONARY ANGIOGRAMS

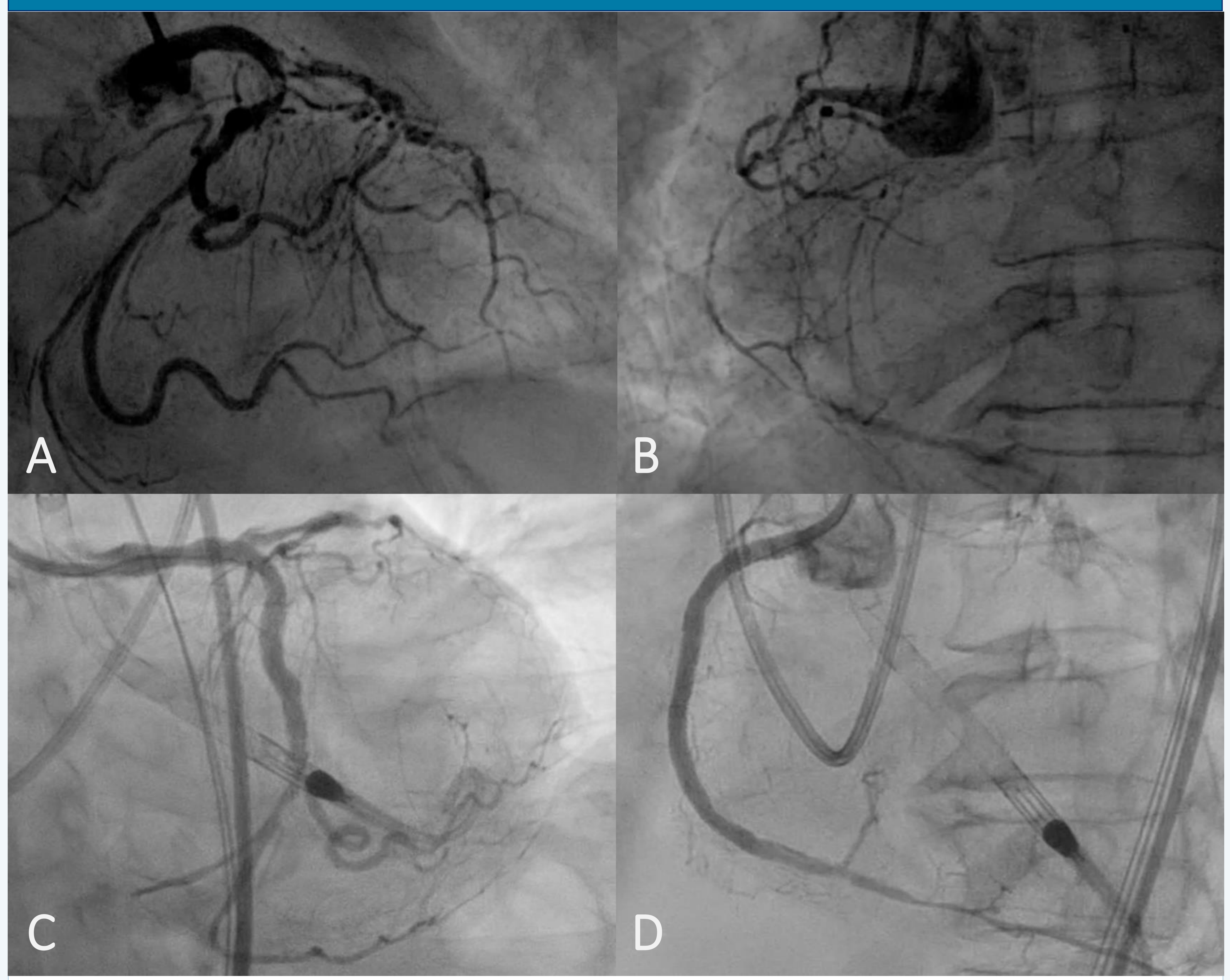


Figure: Composite of coronary angiograms from index emergent coronary angiography (A, B) and final angiograms after high-risk, multivessel PCI (C, D). (A) Initial coronary angiogram (LAO caudal) demonstrating ostial LAD subtotal occlusion with severe distal disease and high grade ostial LCx stenosis. (B) Initial coronary angiogram of R system demonstrating proximal RCA CTO with R>R collaterals. (C) Successful IVUS-guided LM–LCx DES and LAD POBA + DCB. (D) Successful RCA CTO revascularization with overlapping DES.

DISCUSSION & KEY TAKEAWAYS

Triggers for Suspicion

Ischemic ECG changes spanning >1 territory should trigger suspicion for ACS superimposed on chronic obstructive disease, i.e., a double jeopardy MI

Heart Team based care improves efficiency and clinical outcomes

Shock system protocols, early multidisciplinary engagement, and shared decision-making enable appropriate and timely escalation of care

Prompt MCS escalation mitigates clinical deterioration

Early, smoldering AMI-CS was recognized in this case, prompting appropriate initiation of MCS, mitigating subsequent end organ damage

DCB enables metal-sparing intervention at critical bifurcations

Use of DCB in bifurcation lesions offers an alternative modality for PCI to help mitigate risk of jailing branch vessels

Late Revascularization Can Recover Systolic Function

Late presenting ACS still benefits tremendously from revascularization

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